

# Problem Statement

Title :

**InterviewAce AI – An AI-Powered Interview Preparation Platform using IBM watsonx.ai and IBM Granite Foundation Models**

## Problem Statement

Job seekers often struggle to prepare effectively for technical, HR, and behavioral interviews due to limited access to personalized interview practice, realistic questions, and detailed feedback. Most existing interview preparation platforms rely on static question banks or generic assessments that do not adapt to different job roles, company expectations, or candidate skill levels. As a result, candidates lack confidence and are unable to identify areas that require improvement before actual interviews.

## Proposed Solution :

InterviewAce AI is an AI-powered interview preparation platform developed using **Flask** and **IBM watsonx.ai** with **IBM Granite Foundation Models**. The system generates personalized interview questions based on the selected job role, target company, interview type, difficulty level, and candidate skills extracted from an uploaded resume. It evaluates candidate responses using AI and provides scores, strengths, improvement suggestions, and structured feedback to help users improve their interview performance.

The platform also maintains interview history and performance analytics through an interactive dashboard, allowing users to track their progress over multiple practice sessions.

## Objectives :

- Generate personalized interview questions using IBM Granite AI models.
- Support Technical, HR, and Behavioral interview practice.
- Analyze uploaded resumes to personalize interview questions.
- Evaluate answers using AI-generated feedback and scoring.
- Maintain interview history and performance analytics.
- Improve candidate confidence and interview readiness.

## Technologies Used :

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python Flask
- **AI Platform:** IBM watsonx.ai

- **Foundation Model:** IBM Granite (granite-4-h-small)
- **Data Storage:** Browser Local Storage
- **Visualization:** Chart.js

**Expected Outcome :**

InterviewAce AI provides a smart, interactive, and personalized interview preparation experience that helps candidates practice realistic interview scenarios, receive instant AI-powered feedback, monitor their performance over time, and improve their confidence before attending real job interviews.